

Free Software for Studying Science



There are many FOSS tools available in the field of science education. Schools and undergraduate colleges no longer need to buy software for many subjects. Here is a list of free software, almost all of which are available for both the Windows and Linux platforms for college students and staff.

Dr Sastri, an associate professor, wanted his students to do simulation of various topics in physics—but was taken aback by the licence fees for MATLAB, which ran into several lakhs of rupees for just a few licences. He did a search on the Internet, and found a completely free software called Scilab, which could be used for simulations. Scilab is just one example of many such FOSS tools that you should investigate before shelling out large sums of money for commercial software licences. Here is my list—headed, of course, by Scilab.

Scilab: This has been designed for scientific and numerical computations on an open computing platform. It has a command-line facility, and also allows the user to write programs in its own powerful programming language. It has hundreds of mathematical functions, and users can add programs written in other languages, including C and C++, to their program. Scilab has advanced data structures and also allows users to add data types of their own. One of its most important and common applications is simulation.

FreeMat: FreeMat is available for Windows and Linux, and can be used for rapid scientific prototyping and data processing; it is similar to MATLAB.

Maxima: As per Wikipedia, Maxima is a full-featured computer algebra system that specialises in symbolic operations and also offers numerical capabilities. It includes a complete programming language, and is available on both Windows and Linux.

Grace: This is a WYSIWYG 2D graph-plotting tool available on both Windows and Linux. It creates very high quality output, and can be used for both linear and non-linear curve fitting. It can also do Fast Fourier Transforms, integration and differentiation, splines, interpolation and smoothing.

KTEchlab: This is an IDE (Integrated Development Environment) for electronic circuit design and simulation.

Latex: This is used to produce scientific documents, especially where mathematics symbols are involved. Latex can now be used to produce any type of document. It is currently available on most platforms.

Celestia: You can use this 3D astronomy program to view the universe at any point of time and from any place. You can even travel through any part of the universe at the speed of your choice! Celestia displays the orbital path of any object, allows users to orbit around the planets, and track various objects in space. It is available for Linux, Windows and Mac OS X.

Stellarium: This simulates a planetarium. It shows a realistic sky in 3D, just like what you see with the naked eye, binoculars or a telescope, once you enter your positional coordinates. It is available for Windows, Linux and Mac OS.

Chemical Equation Expert: Chemistry professionals and students can use this tool to balance chemical equations and related calculations.

Kalzium: This is a digital replacement for the periodic table.

ACD/ChemSketch: Here is a package that helps you in drawing chemical structures, calculating molecular properties, 2D and 3D structure cleaning and viewing, and in the prediction of logP. It also has a function for naming structures containing less than 50 atoms and 3 rings. It is an excellent tool for chemistry students.

Avogadro: An easy-to-use software for both students and researchers, it features an advanced molecule editor and visualiser designed for use in computational chemistry, molecular modelling, bioinformatics, materials science and related areas. It offers flexible high-quality rendering and a powerful plug-in architecture.

The above list is just a sample of what is available on the Internet. Readers can visit <http://www.bestfreewaredownload.com/> for a comprehensive list of free software along with brief descriptions on each of them. 

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